A New Perspective to Boost Performance: Fairness for Medical Federated Learning

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AI22BTECH11013

13th October 2025

Federated Learning in Healthcare

Federated Learning (FL):

- Enables collaboration across hospitals without sharing raw data.
- Leverages diverse datasets for a stronger global model while preserving privacy.

Why Fairness Matters:

- Standard FL emphasizes only **global performance**.
- Risk: some hospitals benefit less (or even suffer), reducing motivation to participate.
- Fairness ensures all hospitals gain, sustaining collaboration.

Previous Approaches for Fairness in Medical FL

- **Ditto:** Introduced additional objectives and personalized loss per client to improve fairness.
- **FedCE:** Re-weighted clients based on gradient contributions and validation loss to balance global performance.
- **Other Re-weighting Methods:** Used empirical loss or validation metrics to assign overall client weights;
- **limitation:** coarse weighting ignores layer-wise feature differences.

Limitations of Existing Fair FL Methods

Current focus:

- Collaboration fairness (resource allocation).
- Performance fairness (client-level outcomes).

Key limitation:

- Assign a **single weight per client**, ignoring **layer-wise feature shifts**.
- Domain shifts cause layer-level representation differences $\rightarrow$ suboptimal aggregation.

Challenge:

- Need aggregation that accounts for **layer-wise representation differences**.

Motivation & Problem

Motivation:

- Improve fairness in FL for sustainable hospital collaboration.
- Crucial in medical applications with diverse datasets.

Problem:

- Existing fair FL methods **ignore domain shift**.
- Performance disparities reduce incentives for hospital participation.

Challenge:

- How to ensure fairness while accounting for layer-wise feature shifts across hospitals?

Proposed Solution: Fed-LWR

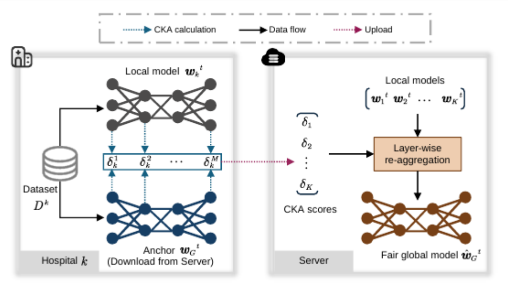
Fed-LWR: Layer-Wise Re-weighting

- Uses the **global model as anchor** and compares local models to it via **layer-wise CKA similarity**, capturing feature representation differences across hospitals.
- Translates these similarity scores into **layer-specific weights**, giving more influence to hospitals whose local features differ significantly from the anchor.
- Performs **layer-wise re-aggregation** using these weights, ensuring the final global model adapts fairly to diverse client data distributions.

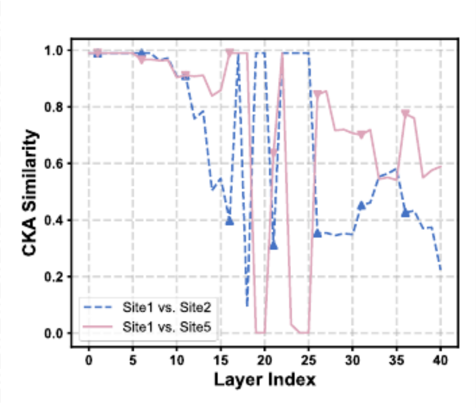
Outcome:

- Better fairness/accuracy trade-off than state-of-the-art.
- Hospitals with weaker performance are not left behind.

Fed-LWR: Illustration and CKA Variation



(a) Overview of Fed-LWR: Layer-wise CKA similarity δ_k is computed between local and anchor models and used to re-aggregate for a fair global model $\hat{w}_t^G$.



(b) CKA variation across layers for two randomly selected client pairs, showing that model differences vary by layer.

Preliminaries: Federated Learning

Setup:

- K hospitals participate in FL via a trusted central server.
- Hospital $k \in [K]$ has private dataset D_k with n_k samples $\{(X_i, Y_i)\}_{i=1}^{n_k}$.
  Each hospital trains a neural network $f = h_M \circ \cdots \circ h_2 \circ h_1$ with M layers.

Federated Learning Objective:

- Minimize local empirical risk L_k for each hospital.
- Update global model via averaging:

$$w_t^G = \frac{1}{K} \sum_{k=1}^{K} w_t^k$$

where w_t^G and w_t^k are global and local model parameters at round t.

Performance Fairness

Definition: Model $\hat{w}$ is more **fair** than w if its performance is more uniform across the K hospitals.

Importance:

- Prevents disproportionate advantage/disadvantage to any hospital.
- Maintains incentives for all hospitals to participate.

Intuition:

- High average accuracy alone is not enough.
- Fairness considers **per-hospital performance balance**.
- The goal is to balance between **high** and **fair** performance.

Dice Score: Evaluation Metric

Definition: Measures overlap between predicted and ground truth segmentation.

$$\text{Dice} = \frac{2|X \cap Y|}{|X| + |Y|}$$

Range: $0 \rightarrow$ no overlap, $1 \rightarrow$ perfect overlap.

Interpretation:

- Sensitive to false positives/negatives.
- Crucial for medical image segmentation (MRI, retinal, etc.)

Advantages: Intuitive, region-focused, widely used.

Use in FL: Evaluates both accuracy and fairness across hospitals.

Hilbert-Schmidt Independence Criterion (HSIC)

Purpose: Measures dependence between two feature sets.

Given: Feature matrices from two layers:

$$U \in \mathbb{R}^{n\times p}, \quad V \in \mathbb{R}^{n\times q}$$

Gram Matrices:

$$K = UU^\top, \quad L = VV^\top$$

Centering Matrix:

$$H = I - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$$

HSIC Calculation:

$$\mathrm{HSIC}(K, L) = \frac{1}{(n-1)^2}\mathrm{tr}(KHLH)$$

Interpretation: Higher HSIC $\rightarrow$ more dependent / similar feature representations.

Centered Kernel Alignment (CKA)

Purpose: Quantifies similarity between two layers' representations using HSIC.

CKA Formula:

$$\mathrm{CKA}(U, V) = \frac{\mathrm{HSIC}(K, L)}{\sqrt{\mathrm{HSIC}(K, K) \cdot \mathrm{HSIC}(L, L)}}$$

Interpretation:

- $\delta = 1$: Layers have identical representations
- $\delta = 0$: Layers are completely dissimilar

Use in Fed-LWR:

- Compute layer-wise similarity between local and global models.
- Lower CKA $\rightarrow$ higher weight for that hospital in re-aggregation.

Algorithm 1: Fed-LWR

Algorithm 1: Fed-LWR

Input: K hospitals, communication rounds T, learning rate η,

Output: $\hat{\boldsymbol{w}}_G^T$

Initialize $\boldsymbol{w}_G^0$ `// ŵ_G^0 = w_G^0`

for *round* $t = 1, 2, ..., T$ **do**

  for *hospital* $k = 1, 2, ..., K$ ***parallelly*** **do**

    $\boldsymbol{w}_k^t \leftarrow \hat{\boldsymbol{w}}_G^{t-1}$

    $\boldsymbol{w}_k^t \leftarrow \boldsymbol{w}_k^t - \eta \nabla \mathcal{L}_k$ `// Local training`

  end

  $\boldsymbol{w}_G^t \leftarrow \frac{1}{K} \sum_{k=1}^{K} \boldsymbol{w}_k^t$ `// Averaging aggregation for the anchor`

  for *hospital* $k = 1, 2, ..., K$ ***parallelly*** **do**

    $\delta_k \leftarrow [\delta_k^1, \ldots, \delta_k^m, \ldots, \delta_k^M]$, $\delta_k^m \leftarrow \mathrm{CKA}(\boldsymbol{U}_k^m, \boldsymbol{V}_k^m)$ `// Eq.` (3)

  end

  $\rho_k \leftarrow [\rho_k^1, \ldots, \rho_k^m, \ldots, \rho_k^M]$, $\rho_k^m \leftarrow \frac{1-\delta_k^m}{\sum_{i=1}^{K}(1-\delta_i^m)}$ `// Eq.` (6)

  $\hat{\boldsymbol{w}}_G^t \leftarrow \{\hat{\boldsymbol{w}}_{G,m}^t\}_{m=1}^{M}$, $\hat{\boldsymbol{w}}_{G,m}^t \leftarrow \sum_{k=1}^{K} \rho_k^m \boldsymbol{w}_{k,m}^t$ `// Eq.` (3)

end

return $\hat{\boldsymbol{w}}_G^T$

Fed-LWR: Representation Difference Estimation (Idea)

Goal: Quantify feature differences between local and global models for each hospital.

Motivation:

- Clients have heterogeneous data distributions $\rightarrow$ feature shift.
- **CKA similarity** can capture representation differences across models.
- Lower similarity $\rightarrow$ global model further from local optimum $\rightarrow$ poorer performance.

Fed-LWR: Representation Difference Estimation (Steps)

After local training in round t, the global model w_t^G (anchor) is sent to each hospital.

For hospital k, compute **CKA similarity** for each layer m:

$$\delta_k^m = \mathrm{CKA}(U_k^m, V_k^m)$$

Construct the layer-wise similarity vector:

$$\delta_k = [\delta_k^1, \delta_k^2, \ldots, \delta_k^M]$$

Fed-LWR: Representation Difference Estimation (Feature Matrices)

Feature matrices used in CKA:

$$U_k^m = h_m \circ h_{m-1} \circ \cdots \circ h_1\left([X_1, \ldots, X_{n_k}],\ w_t^{k,m} \cup \cdots \cup w_t^{k,1}\right)$$

$$V_k^m = h_m \circ h_{m-1} \circ \cdots \circ h_1\left([X_1, \ldots, X_{n_k}],\ w_t^{G,m} \cup \cdots \cup w_t^{G,1}\right)$$

U_k^m: feature matrix from local model k at layer m.

V_k^m: feature matrix from global anchor at layer m.

Fed-LWR: Representation Difference Estimation (Interpretation)

- Higher $\delta_k^m \rightarrow$ local and global representations are similar.
- Lower $\delta_k^m \rightarrow$ greater feature difference, meaning:
  - Global model is further from local optimum.
  - Poorer performance expected for hospital k.
- These similarity scores δ_k^m will later guide **layer-wise weighting in aggregation**.

Fed-LWR: Aggregation Weights Computation

Goal: Assign fair weights to hospitals based on feature differences.

Step 1: Convert CKA similarity to weights for each layer m:

$$\rho_k^m = \frac{1 - \delta_k^m}{\sum_{i=1}^{K}(1 - \delta_i^m)}$$

Interpretation:

- Hospitals with larger feature differences get higher weights.
- Ensures clients with poorer performance influence the global model more fairly.

Fed-LWR: Layer-wise Aggregation & Outcome

Step 2: Re-aggregate each layer of the global model:

$$\hat{w}_t^{G,m} = \sum_{k=1}^{K} \rho_k^m w_t^{k,m}$$

Step 3: Combine layers to obtain fair global model:

$$\hat{w}_t^{G} = \hat{w}_t^{G,1} \cup \cdots \cup \hat{w}_t^{G,M}$$

Outcome:

- Fair global model used as initialization for next local training round.
- Improves performance fairness across hospitals while maintaining global accuracy.

Experiment: Datasets

Dataset	Clients	Image Size	Train/Val/Test Split
ProstateMRI [18]	6	384×384	6:2:2
RIF [29]	4	384×384	4:1 (training split), testing pre-defined

Metrics:

- Dice coefficient for segmentation accuracy
- Standard deviation across clients for performance fairness

Experiment: Baselines

Method	Description
Solo	Clients train independently, no communication
FedAvg [21]	Standard FL with global averaging
q-FedAvg [16]	Client re-weighting via empirical loss
CFFL [20]	Re-allocates server model using validation contribution
CGSV [31]	Cosine gradient Shapley value for contribution estimation
Ditto [14]	Personalized fair FL with client-specific optimization
FedCE [8]	Client re-weighting using gradients and validation loss (Sum./Multi.)

Results: Dice Scores (%) for ProstateMRI and RIF

Method	Sites	Dice Scores (%)	Avg.	Std.
Solo	ProstateMRI: Site1–6	84.41, 85.02, 93.70, 91.06, 89.33, 85.43	88.16	3.46
	RIF: Site1–4	82.01, 74.57, 91.26, 92.56	85.10	7.31
FedAvg [21]	ProstateMRI: Site1–6	85.84, 92.27, 93.58, 87.89, 92.47, 80.66	88.78	4.54
	RIF: Site1–4	82.70, 72.68, 91.19, 91.93	84.63	7.79
q-FedAvg [16]	ProstateMRI: Site1–6	85.84, 94.93, 90.22, 88.53, 90.43, 83.32	88.88	3.67
	RIF: Site1–4	77.83, 80.48, 91.72, 91.62	85.41	6.32
CFFL [20]	ProstateMRI: Site1–6	82.84, 94.49, 89.83, 86.89, 90.83, 82.59	87.91	4.29
	RIF: Site1–4	81.49, 74.97, 91.33, 89.53	84.33	6.55
CGSV [31]	ProstateMRI: Site1–6	82.36, 91.02, 90.14, 87.25, 91.06, 82.71	87.42	3.68
	RIF: Site1–4	78.79, 77.30, 89.77, 91.78	84.41	6.42
Ditto [14]	ProstateMRI: Site1–6	88.15, 93.49, 92.72, 89.99, 92.62, 83.64	90.10	3.42
	RIF: Site1–4	82.11, 78.91, 90.90, 91.95	85.97	5.58
FedCE (Sum.) [8]	ProstateMRI: Site1–6	88.13, 94.02, 91.14, 90.63, 92.27, 87.93	90.69	2.15
	RIF: Site1–4	87.02, 76.60, 90.92, 90.28	86.21	5.73
FedCE (Multi.) [8]	ProstateMRI: Site1–6	89.85, 93.49, 92.21, 90.67, 93.66, 89.53	91.57	1.65
	RIF: Site1–4	86.00, 77.42, 90.69, 89.93	86.01	5.26
Fed-LWR	ProstateMRI: Site1–6	**91.92, 94.73, 93.22, 93.34, 94.06, 92.21**	**93.25**	**0.97**
	RIF: Site1–4	**85.89, 79.43, 92.42, 90.96**	**87.17**	**5.08**

Analytical Studies: Ablation Study

Purpose: Examine the design of Fed-LWR (feature difference measurement and layer-wise aggregation).

Variants:

- **Fed-LWR-v1:** Uses **cosine similarity** instead of CKA to measure feature differences.
- **Fed-LWR-v2:** Uses only the **last layer's CKA similarity** of U-Net encoder for aggregation.

Insights:

- Cosine similarity performs worse than CKA for estimating feature differences.
- Layer-wise aggregation significantly improves fairness and performance.
- Fed-LWR achieves the best balance of accuracy and fairness.

Ablation Study: ProstateMRI Dice Scores (%) (2/2)

Method	Site1	Site2	Site3	Site4	Site5	Site6	Avg.	Std.
FedAvg [21]	85.84	92.27	93.58	87.89	92.47	80.66	88.78	4.54
Fed-LWR-v1	85.76	94.29	94.36	88.51	92.91	86.33	90.36	3.62
Fed-LWR-v2	89.27	94.86	92.99	90.50	93.24	89.32	91.70	2.12
Fed-LWR	**91.92**	**94.73**	**93.22**	**93.34**	**94.06**	**92.21**	**93.25**	**0.97**

Table 3: Dice scores (%) on the testing set of ProstateMRI for different Fed-LWR variants. Avg. and Std. represent the average and standard deviation across clients. Best results are highlighted in bold.

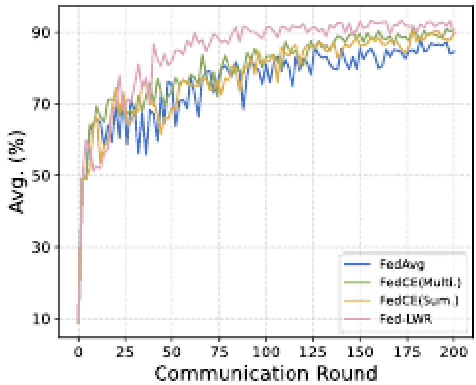
Analytical Studies: Convergence Analysis (1/2)

Objective: Compare convergence of Fed-LWR, FedAvg, and FedCE.

Observations:

- Fed-LWR converges faster; stable after $\sim$90 communication rounds.
- Fairness mechanism does not slow down convergence.

Visualization: Testing performance vs communication rounds



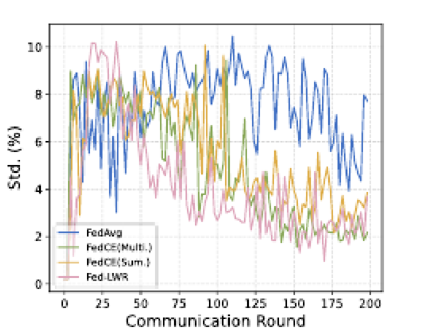
Analytical Studies: Convergence Analysis (2/2)

Observation of Variance:

- Fed-LWR shows larger standard deviation between rounds 20–50 due to rapid convergence.
- This variance reflects the adaptive fairness adjustments rather than instability.

Conclusion:

- Fed-LWR achieves both **fast convergence** and **improved fairness** across clients.
- Layer-wise re-weighting effectively balances accuracy and fairness without sacrificing global performance.



Limitations of Fed-LWR

- **Privacy Exposure:** Layer-wise statistics may still leak information without stronger protection mechanisms.
- **Compute Overhead:** Per-layer similarity estimation (e.g., CKA) increases client and server computation.
- **Limited Generalization:** Designed mainly for segmentation; may not transfer well to classification or detection tasks.
- **Fixed Similarity Metric:** Reliance on a single metric (CKA) may not fully capture domain or distribution shifts.

Hier-Fed-LWR: Motivation & Concept

- **Multi-Level FL Architecture:** Hospitals → Regional aggregators → Global server
- **Enhanced Fairness:** Layer-wise reweighting applied locally and regionally → uniform performance across hospitals
- **Privacy Preservation:** Raw activations never leave hospital; only summaries or divergence metrics sent
- **Communication Efficiency:** Regional aggregation reduces load on global server
- **Scalability:** Hierarchical aggregation supports large federations efficiently

Retinal Fundus Imaging

- **Definition:** A non-invasive imaging technique used to capture the interior surface of the eye, including the retina, optic disc, macula, and blood vessels.
- **Clinical Importance:**
  - Early detection of diabetic retinopathy, glaucoma, AMD, hypertensive retinopathy.
  - Critical for mass screening and longitudinal monitoring of ocular health.
- **Characteristics:**
  - High inter-patient variability due to illumination, pigmentation, device type.
  - Strong domain shifts across hospitals, cameras, and acquisition protocols.
  - Sensitive to noise, blur, and artifacts.
- **Relevance to Federated Learning:**
  - Data is siloed across hospitals—privacy restrictions prevent centralization.
  - Heterogeneity makes model aggregation challenging.
  - Fundus images are ideal for evaluating robustness of FL algorithms.

Summary of Standard FedLwR Behaviour

Training Behaviour

Rapid loss reduction in the first 20–30 rounds across all domains.

Domain3 and Domain4 converge fastest with smooth and stable curves.

Domain1 and Domain2 converge slower and show higher variance.

Domain4 reaches highest training accuracy ($\sim$ 96%), Domain3 $\sim$ 94–95%.

Domain1 and Domain2 eventually exceed 90%, but with significant noise.

Generalization Behaviour

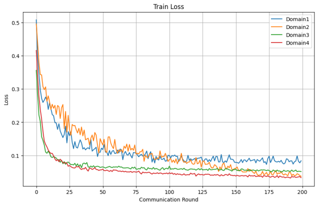
Domain3 and Domain4 show the strongest test performance (90–92%).

Domain1 stabilizes around 82–85%.

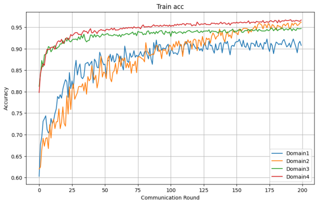
Domain2 suffers from heavy fluctuations and high test loss.

Standard FedLwR struggles on highly shifted domains.

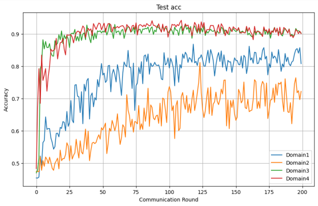
FedLwR Performance Across Domains (All Metrics)



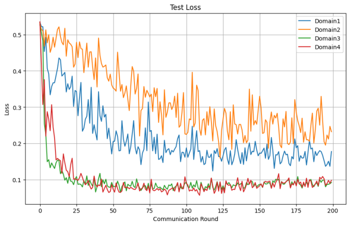
Train Loss



Train Accuracy

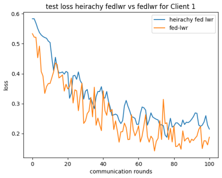


Test Accuracy

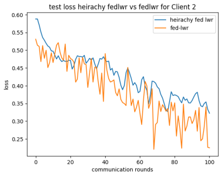


Test Loss

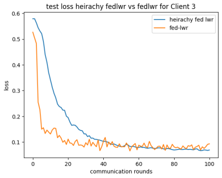
Test Loss and Test Accuracy for Different Domains:Fed-LWR vs Hier-Fed-LWR



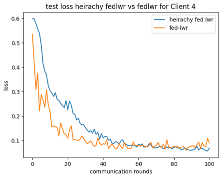
Domain 1



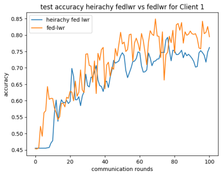
Domain 2



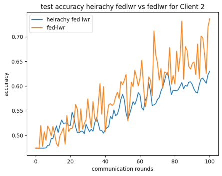
Domain 3



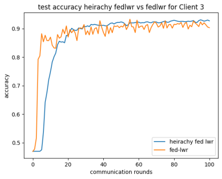
Domain 4



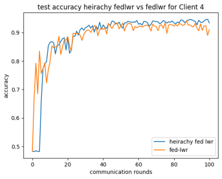
Domain 1



Domain 2



Domain 3



Domain 4

FedLwR vs Hier-FedLwR: Behaviour on Well-Aligned Domains (Clients 3 & 4)

Training Behaviour

Both methods achieve rapid accuracy improvements, reaching 92–95%.

FedLwR converges faster during the initial rounds.

Hier-FedLwR follows closely with smoother accuracy trajectories.

Client 4 shows marginal late-round gains under Hier-FedLwR.

Generalization Behaviour

FedLwR achieves lower test loss and faster convergence.

Hier-FedLwR generalizes similarly but with more stable test curves.

Both methods perform strongly due to well-aligned feature distributions.

FedLwR vs Hier-FedLwR: Behaviour on Moderately Shifted Domain (Client 2)

Training Behaviour

Accuracy trajectories for both methods show substantial fluctuations.

FedLwR exhibits higher accuracy peaks but suffers from instability.

Hier-FedLwR provides smoother and more consistent progress.

Generalization Behaviour

Final test loss values are nearly identical for both approaches.

FedLwR converges faster but oscillates heavily across rounds.

Hier-FedLwR produces more stable test loss curves with reduced variance.

FedLwR vs Hier-FedLwR: Behaviour on Highly Shifted Domain (Client 1)

Training Behaviour

FedLwR reaches higher accuracy peaks but with strong oscillations.

Hier-FedLwR improves gradually with significantly reduced instability.

Generalization Behaviour

FedLwR lowers test loss quickly but exhibits large fluctuations.

Hier-FedLwR achieves smoother convergence and more reliable generalization.

Final loss values for both methods are comparable.

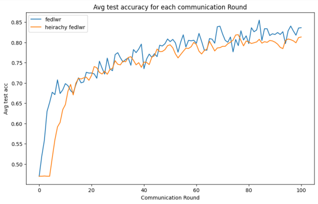
Overall Summary

FedLwR excels on well-aligned domains due to fast, low-loss convergence.

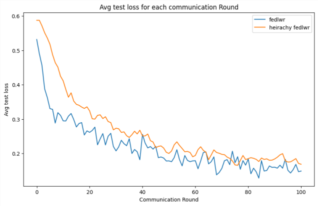
Hier-FedLwR excels on heterogeneous domains by providing improved stability.

Baseline prioritizes speed; Hier-FedLwR prioritizes robustness.

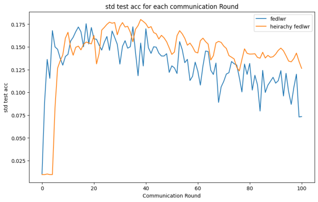
Avg stastics : Fed-LWR vs Heir-Fed-LWR



Avg test acc 1



Avg test loss



Std test acc

FedLwR vs Hier-FedLwR: Accuracy & Stability

- **Lower Variation:** FedLwR shows higher cross-client fluctuation; Hier-FedLwR stabilizes much earlier.
- **Smoothness:** Hier-FedLwR produces cleaner, less oscillatory accuracy curves.
- **Faster Rise (FedLwR):** FedLwR improves accuracy more quickly in early rounds.
- **Final Accuracy:** FedLwR slightly surpasses Hier-FedLwR at convergence.
- **Consistency:** Hier-FedLwR maintains more stable accuracy across heterogeneous clients.

FedLwR vs Hier-FedLwR: Test Loss & Convergence

- **Faster Convergence (FedLwR):** FedLwR reduces test loss more rapidly in early rounds.
- **Lower Final Loss:** FedLwR consistently maintains the lowest loss curve.
- **Stability (Hier-FedLwR):** Hier-FedLwR descends smoothly with fewer fluctuations.
- **Robustness:** Hier-FedLwR offers more reliable behavior under domain shift.
- **Overall Trade-off:** FedLwR = speed & accuracy; Hier-FedLwR = stability & robustness.

FedLwR vs Hier-FedLwR: Final Summary

- **Accuracy:** FedLwR achieves slightly higher final accuracy on well-aligned domains.
- **Stability:** Hier-FedLwR offers smoother and more consistent accuracy/loss trends.
- **Generalization:** FedLwR attains lower test loss; Hier-FedLwR reduces client variance.
- **Robustness:** Hier-FedLwR performs better under strong domain shift, reducing oscillations.
- **Compute Cost:** FedLwR is lighter; Hier-FedLwR can be made cheaper by updating the global model less frequently or weighting only key layers.
- **Trade-off:** FedLwR = **speed + peak accuracy**; Hier-FedLwR = **stability + robustness**.
- **Improvement:** Hier-FedLwR enhances reliability and cross-client consistency at the cost of extra computation.

Thank You

Questions?